

Shipping Wardriving Hardware

Outsourcing Stumbling to Delivery Services

CoD_Segfault

RF Village, DEFCON 31

Who Am I?

A circular profile picture showing a person wearing a white hard hat with a green light on the front, a white face mask, and green hair. The background of the profile picture is dark.

[Edit profile](#)

CoD_Segfault@infosec.exchange
@CoD_Segfault

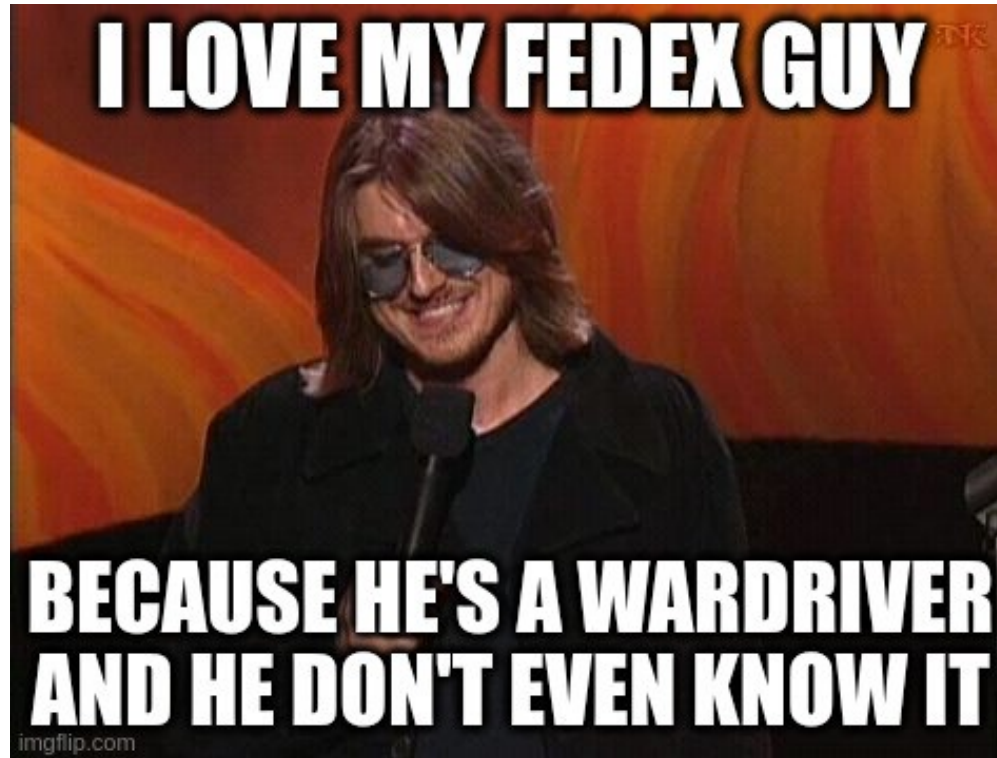
Maker, hacker, tinkerer, things on my head enthusiast. Into electronics, 3D printing, Wi-Fi, and whatever else has my attention.
CoD_Segfault@infosec.exchange

📍 Greater Chicagoland Area 📅 Joined July 2008

202 Following **1,146** Followers

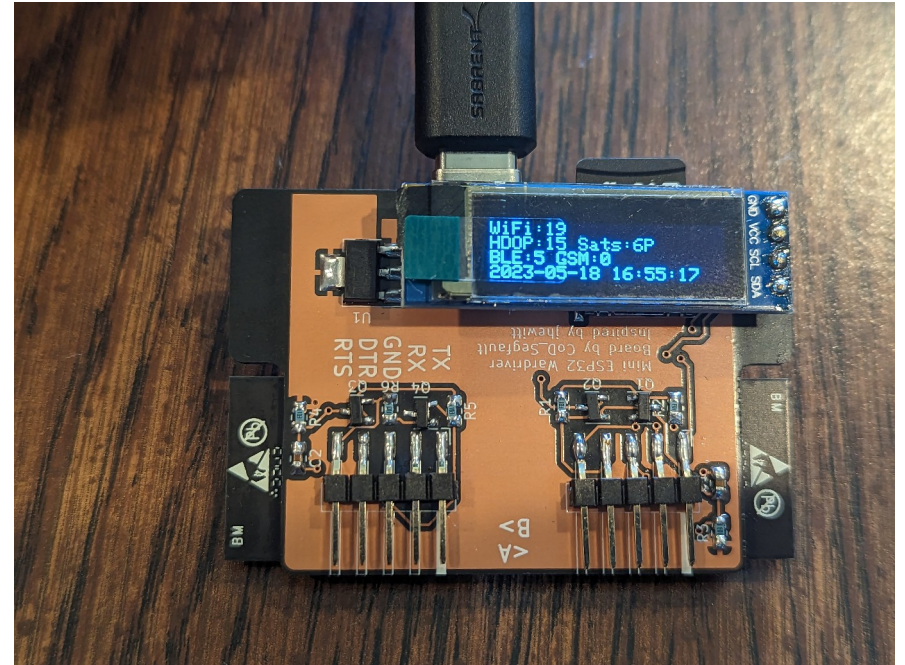
Why Am I Qualified?

- I'm not



Idea

- Determine Safety
- Figure Out Power
- Box and Ship it
- ??????
- Profit: WiGLE Points



Safety Testing

- Ran in Sealed Cardboard Box for 4 Hours
- Used BBQ Thermometer to Monitor
- Only Rose 5°F (~2.7°C) from Ambient



Battery Considerations – Lithium Ion/Polymer



Battery Considerations – Lithium Iron Phosphate

- + Medium Energy Density
- + Safe Operating Conditions
- High Cost
- Shipping Restrictions



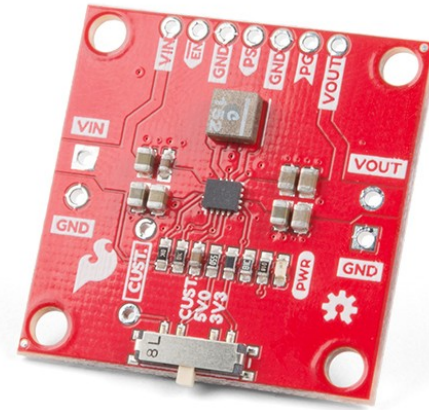
Battery Considerations - Alkaline

- + Easy to Get
- + Cheap
- + Failure Mode is No Power
- + No Shipping Restrictions
- Heavy
- Large
- Low Energy Density



Powering

- First tried cheap Amazon voltage regulators, did not work to advertised specifications.
- Sparkfun Buck-Boost converter (COM-15208)



Box It Up and Wait



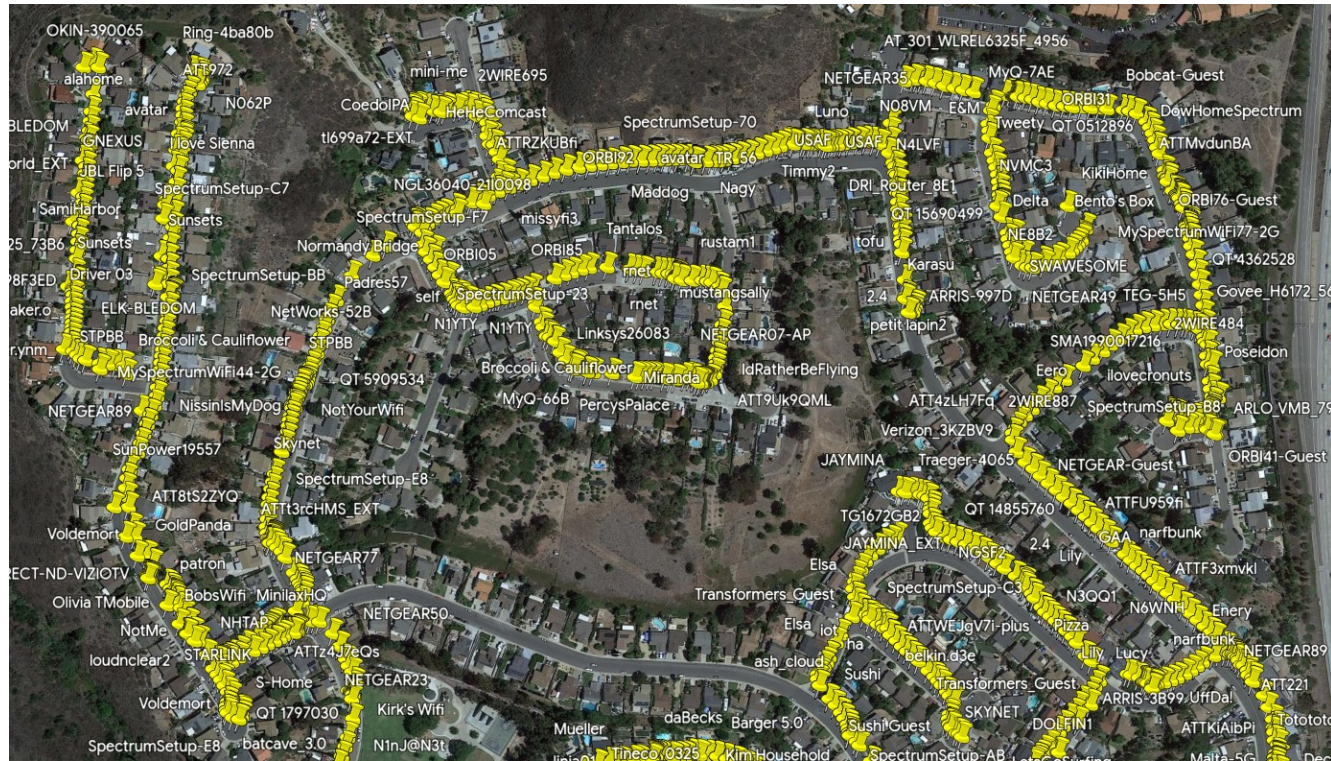
Safe Arrival



Unexpected Results



Good Neighborhood Coverage



Results

Status of Task	Task %age Complete	Seconds to process	Uploaded on Date	File Name (wgle-assigned)	File Size (bytes)	File Lines	WiFi					Cellular					Bluetooth	
							New+ 	New		# in File	New Samples	New+ 	New		# in File	New Samples	New+ 	# in File
Success	100%	20	2023-06-16T22:01:22Z	Segfault7wd3-20.csv	183,007	1,997	5	5	150	151	391	0	0	0	0	0	1,245	1,522
Success	100%	41	2023-06-16T22:01:22Z	Segfault3wd3-16.csv	375,406	4,675	1	463	9	2,474	2,524	0	0	0	0	0	116	2,104
Success	100%	7	2023-06-16T22:01:22Z	Segfault0wd3-13.csv	2,803	31	0	0	0	23	0	0	0	0	0	0	0	5
Success	100%	4	2023-06-16T22:01:22Z	Segfault2wd3-15.csv	4,450	58	0	1	0	36	35	0	0	0	0	0	0	21
Success	100%	10	2023-06-16T22:01:22Z	Segfault4wd3-17.csv	67,238	930	0	96	0	209	229	0	0	0	0	0	0	659
Success	100%	5	2023-06-16T22:01:22Z	Segfault5wd3-18.csv	13,658	185	0	6	0	82	78	0	0	0	0	0	0	95
Success	100%	28	2023-06-16T22:01:22Z	Segfault6wd3-19.csv	188,929	2,400	57	130	59	479	562	0	0	0	0	0	541	1,810
Success	100%	5	2023-06-16T22:01:22Z	Segfault10wd3-23.csv	4,947	49	0	0	18	25	22	0	0	0	0	0	8	23
Success	100%	86	2023-06-16T22:01:22Z	Segfault9wd3-22.csv	1,229,316	12,016	469	470	5,962	5,961	7,866	0	0	0	0	0	3,051	3,974
Success	100%	7	2023-06-16T22:01:22Z	Segfault8wd3-21.csv	25,705	273	18	18	77	93	93	0	0	0	0	0	173	179
Success	100%	176	2023-06-16T22:01:22Z	Segfault1wd3-14.csv	1,977,474	22,924	391	694	3,148	4,513	7,191	0	0	0	0	0	6,054	11,640

13,369 WiFi Observed, 9,221 with Location, 935 New w/Location

Cost Analysis

- Wardriver Board - \$25
 - Voltage Regulator - \$20
 - Batteries - \$25
 - Shipping Cost - \$15
 - Total - \$85
-
- 108 Wifi w/Location per \$1, 11 New WiFi w/Location per \$1

Future Improvements

- Better GPS Antenna
- Multi-Constellation GNSS Module
- Don't Ship to Another Wardriver

Contact

- Twitter: CoD_Segfault
- Mastodon: @CoD_Segfault@infosec.exchange
- Discord: CoD_Segfault